Analyzing the range of angles of a solar panel to detect defective cells, using a UAV

Kyungmin Kim
Software
Sungkyunkwan University
Suwon, Republic of Korea
lufovic77@skku.edu

Yoojin Choi
Computer Science and Engineering
Soongsil University
Seoul, Republic of Korea
yj8677@soongsil.ac.kr

Keonyoung Shim
Software
Sungkyunkwan University
Suwon, Republic of Korea
5kyoungs@g.skku.edu

Hyewon Jeon
Computer and Information Technology
Purdue University
West Lafayette, Indiana, U.S.A
jeon30@purdue.edu

Jane Commerford
Computer and Information Technology
Purdue University
West Lafayette, Indiana, U.S.A
jcommerf@purdue.edu

Eric T. Matson
Computer and Information Technology
Purdue University
West Lafayette, Indiana, U.S.A
ematson@purdue.edu

***Abstract*—Solar energy is now a leading source of renewable energy that can replace fossil fuels. There are many solar farms in the world because it is eco-friendly and can generate electricity semi-permanently. However, these farms are costly with maintenance issues which require high-cost detection. Among various methods for detection, this paper adopts the method of thermal imaging with a UAV to detect defects efficiently. Especially, the power generation efficiency varies significantly depending on the installed angle of the solar panel. For this reason, it is important to consider the angle when the solar panel is installed. This work designs a system analyzing the range of angle between a solar panel and thermal camera to detect defective cells of the solar panel effectively. We developed an algorithm which detects defective solar cells using computer vision and captured out the meaningful outcome by analyzing the result of the experiment. The use of UAV and the thermal camera can reduce the cost and improve the efficiency of maintenance by minimizing human labor. In the future, our algorithm can be automated and scaled to apply to solar farms worldwide.**

Keywords-UAV, Thermal imaging, Fault Detection, PV solar panel, Solar panel tilt angle

I. Introduction

A. General background

These days, the world is using around 101 million barrels of oil per day, the equivalent of approximately $7.4 billion dollars (USD) [1]. An increment in greenhouse gas emissions due to oil consumption leads to climate change and global warming [2]. Renewable energy sources (biofuels, geothermal, solar and wind) play an important role in reducing emissions of carbon dioxide, slowing down global warming and other related issues. Among them, U.S interest in solar energy to replace fossil fuels has been increasing. The Energy Information Administration (EIA) states that the use of Photovoltaic (PV) modules in the US has grown by 59.9% between 2005 and 2015 [3]. Now in 2018, there is enough solar energy produced in a single day in the United States to power up to 11 million houses. Additionally, the capacity of PV is expected to increase by at least two times within 5 years [4].

There are two ways to generate electricity using solar energy: PV power generation using sunlight, or the concentrated solar power generation using solar heat. Of the two, PV power generation is gaining popularity due to the fact that it is an efficient way of generating electricity; additionally, it is getting cheaper due to the development of semiconductor manufacturing technologies [5] [6]. Though the PV power generates energy infinitely, if only one cell is not working among several solar batteries, the generation efficiency will drop drastically and result in irreversible degradation in the near future [7] [8]. If the solar cell is not working normally, the reverse bias happens which blocks the overall current flow due to the reversed direction and stimulates the overheated phenomenon. This phenomenon indicates that the specific cell shows higher temperature when compared to other normal cells. It can be caused by several reasons such as cracks, dirt on the surface or inner circuit short including the reverse bias.

There are several ways to detect defective cells: electricity monitoring of currents or voltages, visual inspection, hotspot detection method using a thermal camera, thermal imaging

978-1-5386-9245-5/19/$31.00 ©2019 IEEE
DOI 10.1109/IRC.2019.00099

with a UAV [9] [10]. The reason why the thermal imaging method was chosen for this research is that this method, unlike the electrical method or visual inspection, is capable of detecting cells damaged by the shunt. A common thermal sensor can effectively detect cell malfunction because a defective cell represents a higher temperature than other nearby cells. Solar panels require continuous management for high generating efficiency, resulting in high cost in maintenance and safety of the employees for large solar panel farms. For example, the largest solar farm area in the world is about around 1,295 hectare (ha), producing enough electricity for 255,000 homes [11]. The maintenance required for a farm of this size would amount to large expenses and safety concerns. For these reasons, many solar panel farms use UAV for maintenance; UAV is light, safe, can gather a lot of data with various sensors and can reduce the maintenance cost to less than 5% of the average annual maintenance cost [12].

Both the tilt angle and the orientation of the solar panel are important factors in determining solar radiation quantity reaching the solar panel. It is obvious that these two factors are directly connected to the efficiency of electricity generation [13]. Consequently, when considering the use of UAVs in solar panel fault detection, these factors must be accounted for. In consideration of these backgrounds, this paper has done the experiments about detecting defective cells of solar panels with different angles between the panel and the ground by using UAV and thermal camera.

B. *Literature Review*

According to Sathyanarayana et al. [7], shading, especially non-uniform shades (such as shadows caused from fallen leaves), make solar panel efficiency significantly decrease lowering to only 2.46% of their potential; it can also cause hot spots on solar panels that cannot be recovered. Wendlandt et al. [14] commented that a hot spot in a solar panel causes the reverse bias, resulting in the leakage-current and harm to the solar panel.

There are many papers about detecting abnormal cells with various proposed methods of detection. One of these methods, introduced by Itako et al. [15], is an integrated system that detects the presence of hotspots. This system used a panel analyzer to produce information. The information was then applied to an algorithm that monitors the voltage and current and then determines the current rate of change as a method of detecting hotspots.

There are some papers that described possible methods to detect defective cells in thermal images. Alsafasfeh et al. [16] used thermal images to detect damaged PV systems. Gao et al. [17] introduced a fault detection method using thermal images and designed a method of taking the images from a moving cart. Similarly, Bellezza Quarter et al. [12] mentioned that using a thermal camera with a UAV provides many benefits including cost efficiency, higher safety ratings than using MAV (Manned Aerial Vehicle) and efficient data collection. Alsafasfeh et al. [5] proposed a solar panel monitoring system that is consisted of one thermal camera (to detect the hot spots) and one CCD camera (to detect external defects such as shadows). These cameras were mounted to a drone, with one ground station for each of the two cameras. Their system allowed for a UAV to survey the PV modules, detect faults and report the longitude and latitudes of the defects. This is a highly efficient method of detecting defects in PV systems. Pia Addabbo et al. [18] also suggested a way to detect defective PV panel by finding PV panel thermal anomalies with RPAS (the Remote Piloted Aircraft System) and GNSS (Global Navigation Satellite Systems). Defective cells are detected through comparison with other cells in a thermal image. Another improvement on the precision of detecting defective PV panels is controlling the way drones take pictures or developing algorithms of the computer vision. Considering these findings, this paper focuses primarily on the possibility of detecting defective PV panels in terms of how they are installed in particularly related to the angle.

In the section 2, the process and the environment of the experiment are described. The section 3 analyzes the outcome of the experiment. In the Conclusion section, a brief summary of this paper and the application of the results that were found in the experiment are provided.

II. Realization

There are 3 subsections in this section consisting of Goals, Approach and Experiment and Evaluation. The experiment's goals are dealt with in the Goals section. The design and experimental environment are described in the Approach section. The steps of the experiment and the analysis of its result are treated in the Experiment and Evaluation section.

A. *Goals*

The first goal is to detect defective cells of a solar panel. To achieve this goal, a thermal camera was used to collect thermal images of the solar panel; Fig. 1 shows the thermal images of the solar panel with defective cells that were emulated using black paper. The second goal is to analyze the range of angles to detect defective cells in a solar panel when using a UAV. To analyze a range of angles between a solar panel and a UAV, the angle between the solar panel and the ground was varied.

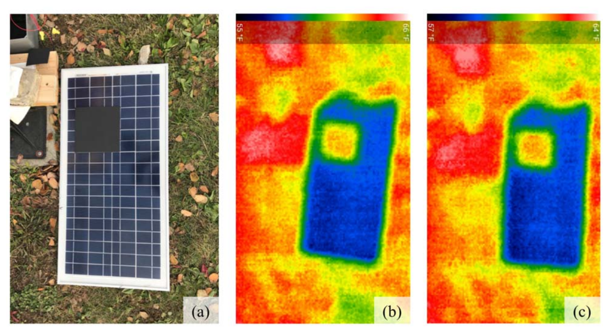
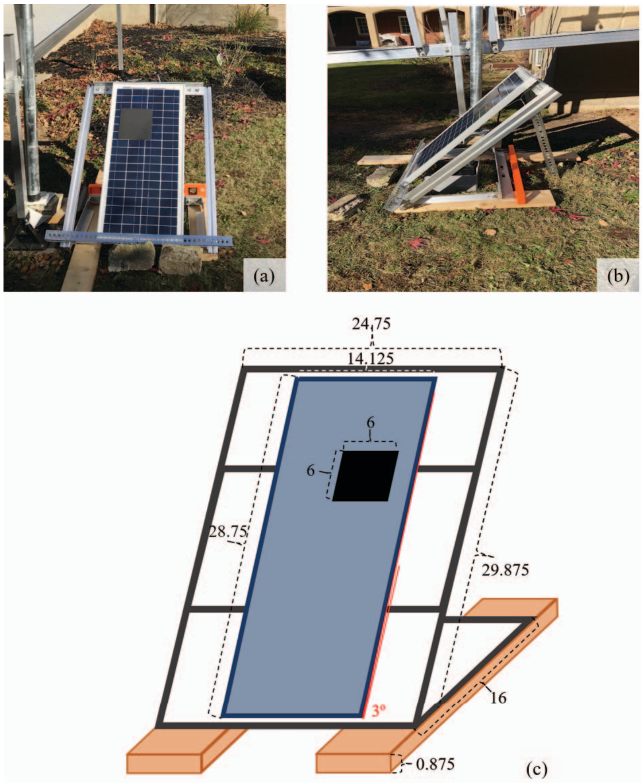


Fig. 1. Detection of defective cells of a solar panel using a thermal camera

B. *Approach*

To define the area of defective cells in a solar panel, a thick black paper was used to cover a certain area of the solar panel. The area of the black paper could work as an abnormal area of a solar panel. The black paper absorbs sunlight well resulting in it getting hotter than the adjacent solar cells. According to Simon and Meyer [8], a defective cell shows a higher temperature than the working (normal) cell does in a thermal image and this is called the hot-spot phenomenon. To differentiate the angle of the solar panel, a solar panel was installed as shown in Fig. 2 to adjust the angle of the solar panel from the ground.



* unit = inch

Fig. 2. Installation of the solar panel

In the experiment, the initial distance between the drone and the solar panel was fixed at 77 inches, the height which made the entire solar panel visible on the screen. Two thermal images of the same solar panel were taken each from 50 and 100 inches away shown in (a) and (b) in Fig. 3. Before comparison, the thermal images, (a) and (b), were cropped to (c) and (d) in Fig. 3. A comparison function, *matchShapes()* in OpenCV, was used to compare the defective area boundary shown in (c) and (d). As a result of the comparison, it showed similarity of 98.52%. With this, it can be inferred the height does not have a significant impact on the detection of an area of defective cells in a solar panel.

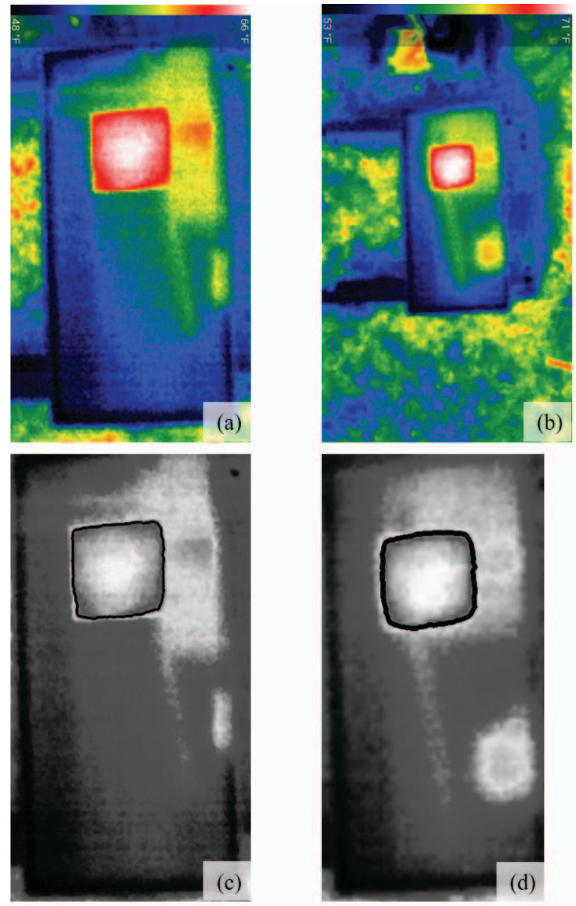


Fig. 3. (a) A thermal image taken 50 inches above the solar panel, (b) A thermal image taken 100 inches above the solar panel, (c) A grayscale image of the solar panel area in (a), (d) A grayscale image of the solar panel area in (b)

In this initial experiment, a pole shown in Fig. 4 was used to hang the drone and simulate a drone-fly. There are several reasons why this was done. First, in this experiment, having an accurate angle between the solar panel and a thermal camera attached to the drone is crucial. By hanging the drone from the pole, errors of angle and height that come from a drone flying remotely can be minimized. Second, by stabilizing the drone rather than flying, it is possible to reduce the impact of external variables such as wind or drone-tilt. Eventually, in the experimental setup, the maximum angle that the drone swing makes is 25.7°. The highest height of the drone was 81 inches and the lowest point of the drone height was 77 inches, so the position error of the drone was just 4 inches vertically, 35.1 inches horizontally.

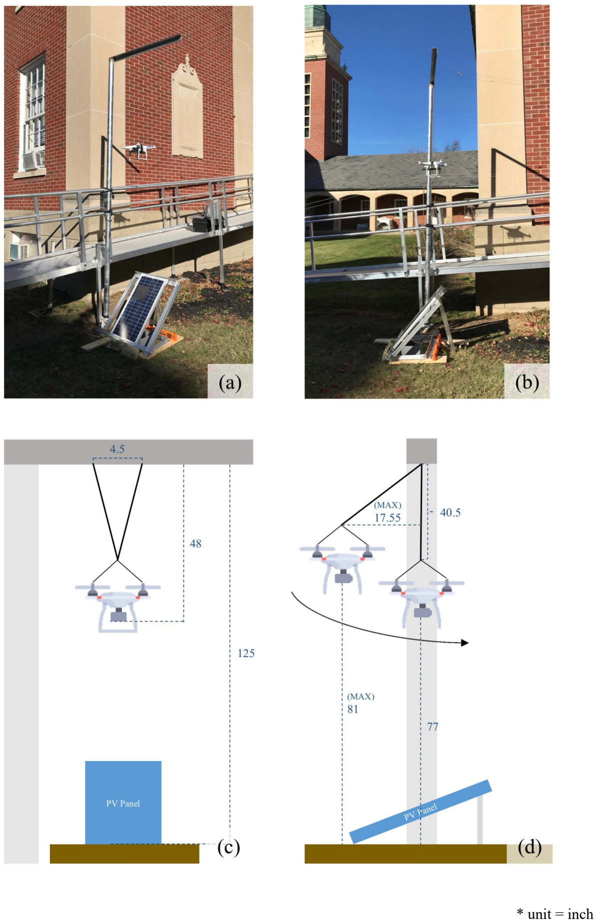


Fig. 4. Installation of the drone and the solar panel

The thermal camera, Seek Thermal Compact Imager for Android [19], was used to record and store thermal images in real-time and have a better network connectivity. The thermal camera was installed to Samsung Galaxy Tab E Lite and attached to the UAV. Once the camera records a video of thermal view of the solar panel, frames were extracted every 0.5 seconds by using FFmpeg, the program that records and transforms many video types, to get thermal images. After the thermal images had been extracted by FFmpeg, an algorithm was used to prepare the dataset for analysis. This algorithm was constructed by using Python and the open source library openCV. The first step in the algorithm was to remove noises from an image because thermal images have fewer pixels than regular images. This is done without blurring the edges of the image so that the contouring can be successful. In order to detect contouring, the image is then converted to a grayscale. A threshold value is selected for converting the grayscale image to black and white. If a pixel is higher than the threshold value it is assigned to white, and if it is lower it is assigned to black. The next process is finding the contours and hierarchy, which is done by connecting all the consecutive points containing the same color or intensity. The contour with the greatest area is defined as the solar panel, and then the contour with the greatest area that lies within the contour of the solar panel is analyzed. Lastly, the contours are then drawn on the top of the original image. Fig. 5 shows this process.

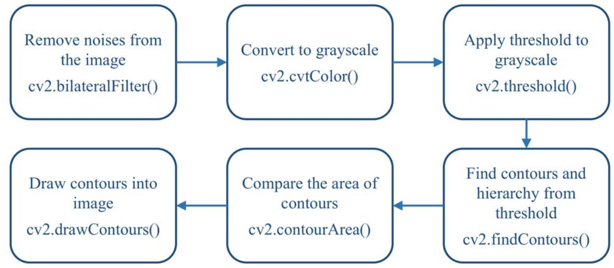


Fig. 5. Flowchart of the detection algorithm

C. *Experiment and Evaluation*

The experiment was done at 13:00 to 15:00 on 11.10.2018 in the afternoon. Weather was sunny, and the temperature was around 30º F.

The experimental procedures are as follows:

- Step 1: Tilt solar panel to the designated angle (ranging from 0º-90º)
- Step 2: Start video taken by remote connection with TeamViewer
- Step 3: Shift the drone across a maximum angle as shown in Fig. 6 over two iterations
- Step 4: Increase the angle of the solar panel with the ground by 10º
- Step 5: Repeat steps 1 - 4

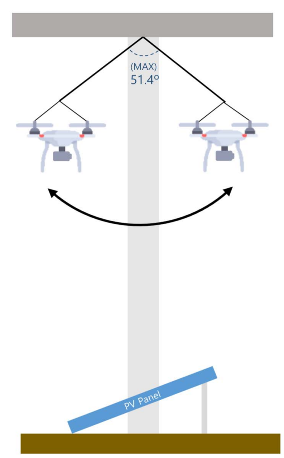
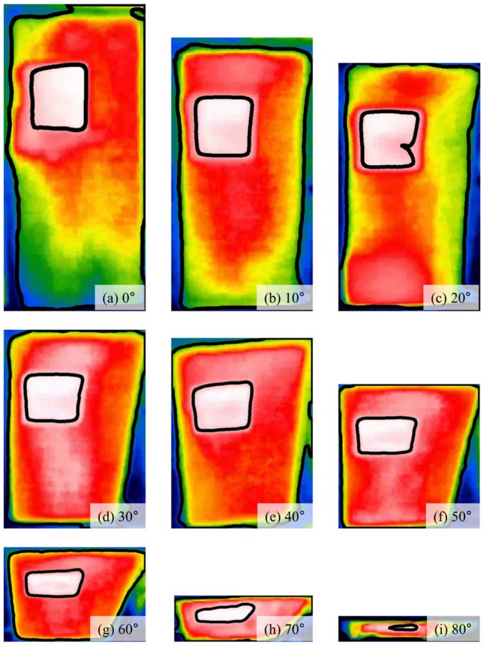


Fig. 6. Drone motion

In order to analyze the videos collected in the experiment, they were processed using the algorithm described previously in the methods section. The area ratio between the defective cells and the solar panel was compared and recorded in the results section.

III. RESULT



* The defective cells were not detected by the thermal camera at 90°.

Fig. 7. Result of detection algorithm

For the 0º to 80º, all the defective cells are detected and for the 90º, nothing was detected, not even the solar panel. The areas of the solar panel and the defective cells are decreased according to increasing the angle. Also, there is a table that analyzes the ratio of the defective cells relative to the solar panel area.

TABLE I. AREA RATIO BETWEEN CONTOURED SOLAR PANEL AND DEFECTIVE CELLS

Degree (°)	***Area of defective cells detected (A) (pixels)***	***Area of Solar panel detected (B) (pixels)***	***Proportion of the area of defective cells in the area of solar panel detected (C[a]) (%)***
0	23121.5	272086.0	8.497864645737009
10	26266.5	275826.0	9.52285136281569
20	24631.0	261112.0	9.43311682343209
30	26436.5	264889.0	9.980218129103134
40	28854.0	278287.0	10.368432589377154
50	22775.0	215869.0	10.550380091629647
60	19849.5	154166.0	12.87540702878715
70	12188.5	87231.0	14.886172109724225
80	1290.0	37834.5	3.40958648852238
90	N/A	N/A	N/A

a. C = A / B * 100

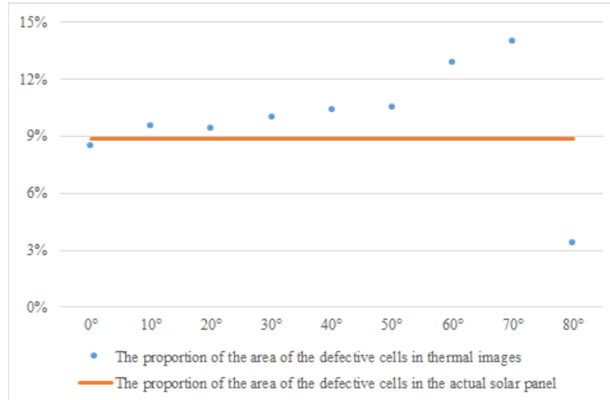


Fig. 8. Installation of the drone and the solar panel

The actual area of the solar panel is 406.09375 in^2 and the actual area of the defective cells is 36 in^2. Ergo, the portion of the defective cell in the solar panel is 8.8649481%.

The result of the experiment proved that the area of the defective cells in the solar panel is detected from 0º to 80º; furthermore, the greater the angle of the solar panel, the larger the ratio of the area of defective cells in the area of solar panel. In the case of 0º there less than a 1% difference between the measured value and the actual value. There is a difference of approximately 1% with an angle of 10º - 30º, about 2% for 40º - 50º, about 4% for 60º, about 6% for 70º, and about 5% for 80º. This indicates that the detection of the defective cells is distorted by increasing the angle of the solar panel. Therefore, the error due to the installation angle of the solar panel should be considered in detection of the defective cells.

IV. CONCLUSIONS

In order to detect defective solar cells, the experiment was conducted with the thermal camera attached to the UAV. In the experiment above, the solar panel was tilted in various angles to detect defective cells and the result of the work is analyzed with computer vision. In the result, the area of defective cells was detected in the range of 0º to 80º. However, as the angle of the solar panel increased, the error of the result became large. This means that distortion of the detection could appear depending on the angle. From this result, the relationship between the angle of the solar panel and the detection of the defective cells is defined. However, the angle of the solar panel cannot be altered because it is commonly installed at the optimal angle for sunlight exposure. Therefore, considering the angle between the solar panel and the thermal camera with the paper result, the error in the fault detection method will be dropped. It will contribute to the reduction of solar panel maintenance cost when using the thermal imaging with a UAV. Furthermore, taking these all into account, the defective solar cells in large solar farms can be easily detected by mapping the small and high-resolution thermal images taken from the autonomous drone flying with GPS.

ACKNOWLEDGMENT

This research was supported by the MSIP(Ministry of Science, ICT & Future Planning), Korea, under the National Program for Excellence in SW) (2018-0-00209-001) supervised by the IITP(Institute for Information & communications Technology Promotion.

REFERENCES

[1] "Short-Term Energy Outlook - U.S. Energy Information Administration (EIA)", EIA, 2018. [Online]. Available: https://www.eia.gov/forecasts/steo/report/global_oil.cfm. [Accessed: 30- Oct- 2018].

[2] "Causes of Climate Change | Climate Change Science | US EPA", EPA. [Online]. Available: https://19january2017snapshot.epa.gov/climate-change-science/causes-climate-change_.html. [Accessed: 30- Oct- 2018].

[3] P. Beiter, T. Tian, M. Meshek and A. Hicks, 2015 Renewable Energy Data Book. Washington, D.C.: United States. Dept. of Energy. Office of Energy Efficiency and Renewable Energy, 2016.

[4] "U.S. Solar Market Insight | SEIA", SEIA, 2018. [Online]. Available: https://www.seia.org/us-solar-market-insight. [Accessed: 30- Oct- 2018].

[5] Alsafasfeh, Moath et al. "Unsupervised Fault Detection and Analysis for Large Photovoltaic Systems Using Drones and Machine Vision." Energies 11.9 (2018): 2252.

[6] Zsiborács, Henrik et al. "Technical-economic study of cooled crystalline solar modules." Solar Energy 140 (2016): 227-235.

[7] Sathyanarayana, P. et al. "Effect of shading on the performance of solar PV panel." Energy and Power 5.1A (2015): 1-4.

[8] Simon, Michael, and Edson L. Meyer. "Detection and analysis of hot-spot formation in solar cells." Solar Energy Materials and Solar Cells 94.2 (2010): 106-113.

[9] D. Kim, J. Youn and C. Kim, "Automatic Photovoltaic Panel Area Extraction from UAV Thermal Infrared Images", Journal of the Korean Society of Surveying, Geodesy, Photogrammetry and Cartography, vol. 34, no. 6, pp. 559-568, 2016.

[10] Tsanakas, John A., Long Ha, and Claudia Buerhop. "Faults and infrared thermographic diagnosis in operating c-Si photovoltaic modules: A review of research and future challenges." Renewable and sustainable energy reviews 62 (2016): 695-709.

[11] "These Are The Biggest Solar Power Plants In America | Alba Energy", Alba Energy, 2018. [Online]. Available: https://albaenergy.com/biggest-solar-power-plants-in-america/. [Accessed: 30- Aug- 2018].

[12] Quater, Paolo Bellezza et al. "Light Unmanned Aerial Vehicles (UAVs) for cooperative inspection of PV plants." IEEE Journal of Photovoltaics 4.4 (2014): 1107-1113.

[13] Benghanem, M. "Optimization of tilt angle for solar panel: Case study for Madinah, Saudi Arabia." Applied Energy 88.4 (2011): 1427-1433.

[14] Wendlandt, Stefan et al. "Hot spot risk analysis on silicon cell modules." 25th European Photovoltaic Solar Energy Conference, Valencia, Spain. 2010.

[15] Itako, Kazutaka et al. "Development of real time hotspot detection system using scan-method for PV generation system." Industrial Electronics Society, IECON 2015-41st Annual Conference of the IEEE. IEEE, 2015.

[16] M. Alsafasfeh, I. Abdel-Qader and B. Bazuin, "Fault detection in photovoltaic system using SLIC and thermal images," 2017 8th International Conference on Information Technology (ICIT), Amman, 2017, pp. 672-676

[17] Gao, Xiang et al. "Automatic solar panel recognition and defect detection using infrared imaging." Automatic Target Recognition XXV. Vol. 9476. International Society for Optics and Photonics, 2015.

[18] Addabbo, Pia et al. "A UAV infrared measurement approach for defect detection in photovoltaic plants." Metrology for AeroSpace (MetroAeroSpace), 2017 IEEE International Workshop on. IEEE, 2017.

[19] "Seek Thermal Compact Specifications", Thermal.com. [Online]. Available: https://www.thermal.com/uploads/1/0/1/3/101388544/compact-sellsheet-usa_web.pdf. [Accessed: 06- Nov- 2018].